Spawning induction by thermal shock of the Mediterranean mussel (*Mytilus galloprovincialis*), and general embryology.

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– PROTOCOL –

Spawning by thermal shock

> *Mussels can be spawned either collectively in a container or individually in cups/beakers of around 250 mL. Spawning mussels in a communal container will significantly reduce the hands-on time, but will increase the probability of multiple crosses if adequate caution is not exercised. If strictly individual crosses are necessary, consider spawning mussels individually.*

1. Pre-heat two different stocks of filtered sea water (FSW) or filtered artificial sea water (FASW) at ∼14 °C (cool water) and ∼26 °C (warm water), respectively.

> *The difference in temperature between cool and warm water should be ∼10 °C.*
>
> *If using FASW made from tap water, a conditioner for marine aquarium shouuld be used.*

2. Collect adult mussels and either refrigerate them at 4 °C overnight or place them in ice for 30–60 min.
3. *(Optional)* Acclimate mussels in FSW/FASW at 16 °C for 30–60 min.
4. Place mussels in the cool water for about 20–30 min.
5. Discard the cool water and replace with the warm water. Let mussels rest for 20–30 min.
6. Check whether any spawning has already occured. If keeping mussels in a communal container, promptly isolate individuals in separate cups/beakers as soon as they start spawning, taking care of carefully wash them with FSW/FASW and air dry. If no spawning can be observed, repeat from step 4.

> *Mussels tend to close their valves when placed in the cool water, so spawning often happens more intensely in the warm water.*
>
> *Spawning can be encouraged/enhanced by gently tapping the container(s).*

7. During spawning, male mussels are recognised because water turns milky (sperm), while female mussels are recognised because an orange-ish material (eggs) deposits at the bottom of the container.

Fertilization and embryo rearing

1. After 1 h since spawning started, collect the desired pair(s) of females and males.

> *If performing multiple crosses, gametes from multiple females and males can be mixed together.*

2. Filter the eff suspension using a 75 µm over a 30 µm nylon mesh (egg size is ∼40–50 µm), in order to remove debris and concentrate gametes.

3. Diluite eggs to 1 L of FSW/FASW and quantify them, by counting in three replicates. Age eggs in FSW/FASW for 45–60 min until they acquire a round shape.

4. If labelling sperm mitochondria, add 1 µL of MitoTracker™ Red CMXRos 1 mM stock solution to 2 mL of sperm suspension (working concentration of 500 nM). If embryos do not segment properly, add 1 µL of MitoTracker™ Red CMXRos 1 mM stock solution to 10 mL of sperm suspension (working concentration of 100 nM). Incubate for 20 min at room temperature taking care to protect from light.

> *From this step onward, if sperm are stained with MitoTracker, keep samples in the dark.*

5. Add the (stained) sperm suspension into the egg suspension to reach an egg sperm ratio of 1:10. Stir vigorously. Check under a microscope for proper ratio. Eggs can be concentrated using a nylon mesh before this step.

> *Be determined and fast at this step, in order to have a synchronised fertilization of most eggs. If keeping adding sperm over several minutes, embryo development will probably result to be strongly asynchronous.*
>
> *Do not exceed the egg/sperm ratio of 1:10, otherwise polyploidisation of embryos will occur frequently.*

6. Leave sperm to fertilize eggs for ∼30 min. After this time, eggs can be *optionally* washed with FSW/FASW to remove excess sperm and avoid polyspermy.

7. Stir vigorously and check for fertilization success by assessing the presence of polar bodies under a microscope.

8. Keeping in mind the estimated initial concentration of eggs, aliquote the embryos to a suitable growing container (e.g., multi-well plates or cell culture flasks) and add FSW/FASW to reach a final concentration of ∼200–250 embryos/mL.

9. Keep embryo preferably at 16 ± 1 °C.

10. Before 48 hpf, embryo do not need to be fed or oxygenated.

11. *(Optional)* Change water 24 hpf by filtering embryos with a >30 µm mesh.

12. If rearing larvae after 48 hpf (D-veliger stage), change water and feed them every 2–3 d. During early stages, larvae can be fed with the flagellate *Isochrysis galbana* (cell size ∼5 µm) at a concentration of ∼100000 cells/mL, or with the diatom *Chaetocerus muelleri* (cell size of ∼5–10 µm) at a concentration of ∼75000 cells/mL. If larvae need to be reared for longer periods, follow the guidelines by Helm et al., 2004.

Embryo collection and fixation

> *Preferably execute this part of the protocol under constant and gentle rocking (∼600 rpm).*
>
> *If not using sieving baskets, spin samples between each step for 2–3 min at 5000 rpm.*
>
> *If sperm is stained with MitoTracker, keep samples in the dark.*

1. Collect the desired amount of samples and immerse in 3.2 % PFA in 1× PBS fixative.

> *Do not exceed the ratio of samples/fixative of 1:3.*
>
> *When processing many samples at the same time, add few drops of the fixative solution directly to the embryos/larvae. This will prevent developmental unpairing. Additionally, this helps when dealing with samples older than 24 hpf, as they possess cilia and are highly motile (they can not be pelleted by spinning).*

2. Incubate embryos in the fixative solution overnight at 4 °C or for 1–2 h at room temperature.
3. Discard the fixative solution and wash samples 3 × 20 min in 1× PBS/PBST at room temperature.
4. Discard the 1× PBS/PBST and progressively dehydrate embryos in 25, 50 and 75 % methanol in 1× PBS for 20–30 min at room temperature.

> *Samples can be stored long term in absolute methanol at –20 °C.*

– RECIPES –

1× PBS with 0.1 % Tween 20 (PBST or PTw)

Compound	Quantity	Molar mass (g/mol)	Final concentration (for 50 mL)
Tween 20	50.0 µL	1,227.54	0.1 %
20× PBS	5.0 mL	–	1×
Ultrapure water	–	–	–

1. Add 5 mL of 20× PBS to a graduated cylinder.
2. Add 50 µL of Tween 20.
3. Fill up to 50 mL with ultrapure water.

3.2 % paraformaldehyde (PFA) fixative

Compound	Quantity	Molar mass (g/mol)	Final concentration (for 100 mL)
Paraformaldehyde	3.2000 g	–	3.2 %
1 M NaOH	as needed	–	–
1 M HCl	as needed	–	–
1× PBS	–	–	–

1. Under a ventilated hood, add 80 mL of 1× PBS into a glass beaker and preheat at 60 °C (mind that the solution does not boil);
2. while stirring, careefully add 3.2 g of PFA;
3. gently raise the pH by adding 1 M NaOH dropwise with a glass pipette: this will speed up PFA solubilization;
4. when PFA is fully dissolved and the solution is clear, adjust the volume to 100 mL;
5. *(Optional)* Filter the solution with filter paper disks;
6. if necessary, adjust the pH to 7 by adding 1 M HCl dropwise with a glass pipette;
7. the fixative solution can be stored for short-term usage at 4 °C or for long-term usage at −20 °C, possibly protected from light.

– RESOURCES –

- Helm, M. M., Bourne, N., & Lovatelli, A. (2004). The hatchery culture of bivalves: a practical manual. FAO.
- Hua, Z., Churches, N., & Nuzhdin, S. V. (2023). Balancing selection and candidate loci for survival and growth during larval development in the Mediterranean mussel, *Mytilus galloprovincialis*. *G3: Genes, Genomes, Genetics*, *13*(7), jkad103. 10.1093/g3journal/jkad103
- Miglioli, A., Dumollard, R., Balbi, T., Besnardeau, L., & Canesi, L. (2019). Characterization of the main steps in first shell formation in *Mytilus galloprovincialis*: possible role of tyrosinase. *Proceedings of the Royal Society B: Biological Sciences*, *286*(1916). 10.1098/rspb.2019.2043
- Rafiq, A., Capolupo, M., Addesse, G., Valbonesi, P., & Fabbri, E. (2023). Antidepressants and their metabolites primarily affect lysosomal functions in the marine mussel, *Mytilus galloprovincialis*. *Science of the Total Environment*, *903*, 166078.. 10.1016/j.scitotenv.2023.166078